Viscosity of a Non-Newtonian Fluid

Pre-Lab Plan

Unit Operations Lab # 6

11/11/2025

Group #2, Tuesday, 9 AM Section

Team Members:

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1. Experimental objective

The objective of this experiment is to analyze the flow behavior of a non-Newtonian fluid (Carbopol solution) using a capillary viscometer by measuring the pressure drop and volumetric flow rate through capillary tubes of different dimensions. The experiment aims to evaluate how apparent viscosity changes with shear rate by determining the flow behavior index(n) and consistency coefficient (m) from the Ostwald-de Waale model. This is done by conducting runs at various pressure differentials and comparing the experimental results with theoretical correlations to determine whether the fluid is pseudoplastic or dilatant.

2. Experimental

This experiment is conducted using a capillary tube viscometer to describe a Carbopol solution's non-Newtonian behavior under different operating conditions. A 1.0wt% Carbopol 934 solution is prepared in warm water and neutralized to pH 5-6. This is done by taking the volumetric flowrates at 3 pressure differentials for each capillary tube and repeated with different L/D ratios to assess how tube geometry affects the fluid's flow characteristics.

a. Apparatus

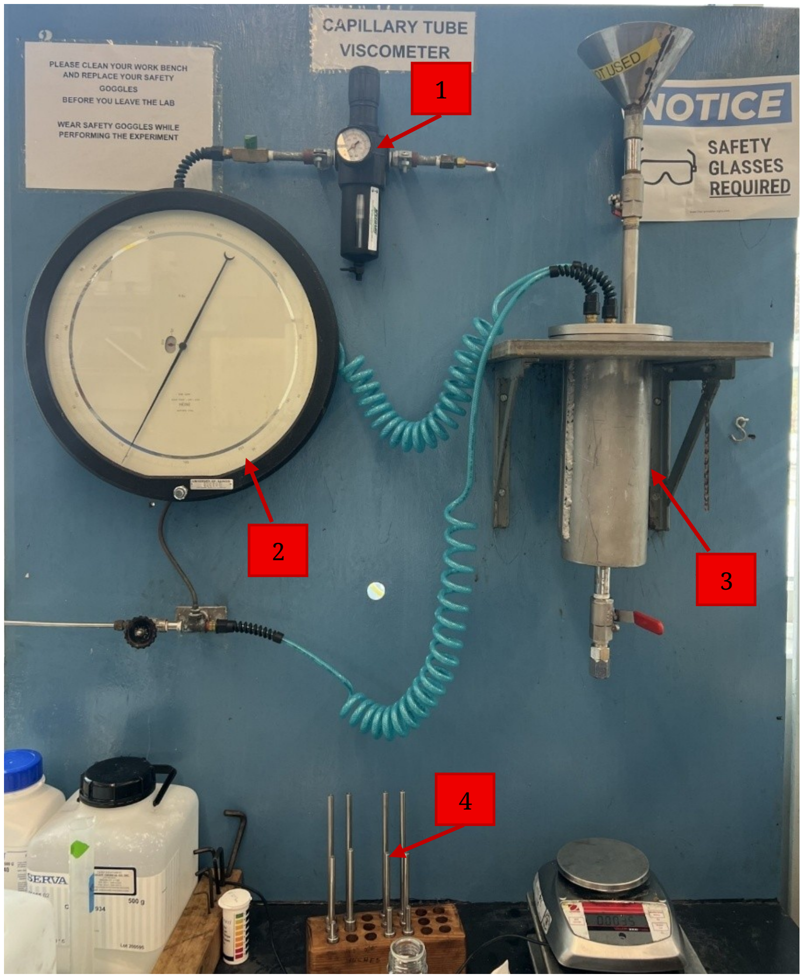

Figure 1: *Apparatus for Non-Newtonian Fluids Experiment*

Table 1: Apparatus Components for Non-Newtonian Fluids Experiment		
No.	**Component**	**Description**
1.	Pressure Regulator	Used to regulate the pressure of the vessel
2.	Pressure Gauge	Reads the pressure controlled

		by the pressure regulator.
3.	Pressurization Vessel	Filled with fluid for which viscosity will be pressurized and measured.
4.	Capillary Tube	Used to measure the viscosity of the fluid.

b. Materials and supplies

Table 2: Materials and Supplies for Non-Newtonian Fluids Experiment

Materials & Supplies	**Description**
Carbopol	Polymer is used to make viscous solution.
Grease	Used to vacuum seal the pressurizing vessel.
Allen Wrench and screws	Used to screw the top of the pressurizing vessel.
Wrench	Used to tighten the capillary tubes at the bottom of the vessel.
Scale	Used to weigh the solution for density calculations.
pH Strips	Used to measure the pH of the viscous solution.
Stand Mixer	Used to mix the solution until desired pH is reached.
Plastic Containers	To store the solution.
Spatula	To handle the viscous solution.
Water	Used to make the solution and clean the apparatus after the experiment is done.
Air	Used to pressurize the vessel.
NaOH Solution	Used to make the solution.
Beakers	For preparing the Carbopol solution.

Graduated cylinders	To measure the amount of NaOH necessary for the solution.
Lab Jack	To aid in mixing the viscous solution.
Measuring Tape	To measure the length of the capillary tubes.
Gloves	To prevent contact with the solution.
Safety Goggles	To prevent the solution from contacting the eyes.

c. Experimental plan

1. Prepare approximately 2 liters of a 1.0wt% Carbopol 934 solution in warm water using a power mixer until completely dissolved.
2. Neutralize the solution by slowly adding 1 N NaOH to adjust the pH to between 5-6, forming a gel-like consistency. Record the final pH value.
3. To set up the apparatus, connect the capillary tube to the pressurization vessel.
4. Fill the vessel with the prepared Carbopol solution.
5. Ensure that the vent valve is opened before filling and closed last during pressurization to maintain safety.
6. Start the experiment by applying compressed air to the vessel and adjust the regulator to maintain a constant inlet pressure.
7. Measure the volumetric flow rate (Q) at several pressure differentials (ΔP) for each capillary tube.
8. Use at least three data points per run. The maximum pressure used should be is 60 psig
9. Repeat the experiment for tubes with different L/D ratios to observe the effect of geometry on measured parameters.

3. Project safety evaluation

Table 3. Safety Evaluation		
Potential electrical hazards?	No	If yes, identify conditions:
Potential mechanical hazards?	Yes	If yes, identify conditions: Yes, the stand mixer poses a mechanical hazard as it spins quickly to homogenize the solution.
Potential pressure hazards?	Yes	If yes, identify conditions: The apparatus will be performing highly pressurized
Potential temperature hazards?	No	If yes, identify conditions:
Hazardous/toxic chemicals involved?	Yes	If yes, hazards involved: Carbopol and NaOH are both toxic chemicals that must be handled with proper PPE
Gloves required?	Yes	If yes, specify type: Nitrile Gloves
MSDS reviewed by all team members?	Yes	List of MSDS sheets reviewed: NaOH Air Water Carbopol
Process Parameters	*Max Expected*	*List location and possible hazards and precautions*
Temperature (ºC)	No	
Pressure (psig)	50-70	The apparatus will operate at high pressures, ranging from 50-70 psig
Safety Controls	Yes	The apparatus is equipped with a pressure relief valve.

I have read relevant background material for the Unit Operations Laboratory entitled:

"Viscosity of a Non-Newtonian Fluid"

and understand the hazards associated with conducting this experiment. I have planned my experimental work in accordance with standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

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